

Toward Hybridized Parameter Estimation for Boundary-Based Control of Synchronous Machines

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Abstract— Hybridized parameter estimation methods that fuse deterministic accuracy with data-driven adaptability remain rare in practice. Deterministic techniques provide physically interpretable parameters but struggle to capture nonlinear, time-varying effects, while heuristic approaches adapt well to changing conditions but often lack transparency and machine invariance. This work proposes a hybridized estimation framework that unites these strengths to generate constraint-aware, behavior-based parameters directly linked to Efficiency Maximization (EM), Optimized Capability (OC), and Protection Resilience (PR) limits. These parameters quantify deviations between idealized machine models and real-world performance under disturbances, enabling real-time forecasting of operational trajectories. Embedded within a boundary-based control scheme, the approach supports adaptive setpoint optimization via nonlinear control, maintaining safe, reliable, and efficient operation across machine types, ratings, and modes. The proposed framework provides a scalable foundation for advanced control architectures in synchronous machine applications.

Index Terms-- Adaptive Control, Hybrid Systems, Parameter Estimation, Power System Protection, Synchronous Machines.

I. INTRODUCTION

Parameter estimation for synchronous machines is a long-standing research topic with foundational principles introduced by notable pioneers [1]–[5]. The latter half of the twentieth century saw the adoption of standardized offline testing methods. IEEE 115 [6] formalized approaches to derive physical synchronous machine parameters. These are strictly offline methods, which rely on manual testing, and assume small-signal linearity. They are unable to represent actual machine behavior since these methods do not account for nonlinearities such as aging or temperature.

By the early 2000s, nonlinear optimization methods emerged, allowing inclusion of physical and operational limits but risking local minima and long computation times. Evolutionary algorithms like Genetic Algorithm (GA) and Particle Swarm Optimization (PSO) search wider spaces to avoid local minima but converge slowly [7], [8]. Today, AI-based heuristics, such as Artificial Neural Networks (ANNs),

can model nonlinear, high-dimensional systems but require extensive training and often lack physical interpretability unless integrated with deterministic methods [9].

Several reviews of estimation techniques exist. Ghnomi *et al.* [10] found no universal method in pre-2007 work. Berdin *et al.* [11] reviewed methods spanning 2001–2019, noting that data-driven and Newton-based approaches both achieve high accuracy. They recommended pairing heuristic approaches with deterministic methods to address limited training data. They observed that many techniques still lack real-time capability due to reliance on full-generator models.

Thus, most methods fail to tie synchronous machine parameters to efficiency, protection, and capability limits, preventing dynamic boundary definition. This leaves these parameters prone to human error and parameter drift.

This paper presents a framework and an illustrative comparison rather than a full controller implementation. Boundary-based control is outlined conceptually, with controller realization and experimental validation identified as future work.

II. SHORTFALLS OF CONTEMPORARY PARAMETER ESTIMATION TECHNIQUES

Contemporary parameter estimation techniques rely heavily on human input, making them vulnerable to both random and systematic errors [10], [6]. Such errors compromise model accuracy and, by extension, the control and protection schemes that depend on these models. Even advanced deterministic approaches, such as Newton-based methods, are not immune.

Heuristic or data-driven methods can replicate complex mathematical processes and, in some cases, produce faster or more accurate results [9], [12]–[14]. However, they still require supervised training, careful architecture selection, and ongoing validation against real-world data. This dependency limits their autonomy and adaptability.

Most existing methods treat parameter estimation as a static exercise. They apply a disturbance, record the response, and estimate the parameters [6], [18], [19] without explaining

why specific values occur or how they vary under changing conditions. This lack of interpretability means that efficiency, capability, protection, and stability issues remain undiagnosed. Moreover, no current approach fully captures real-time machine dynamics, predicts parameter evolution under operational changes, or embeds constraint-awareness into the estimation process [15]–[17], [20]–[23].

A genuinely intelligent, real-time estimation framework must therefore combine deterministic accuracy with heuristic adaptability, enabling the capture of dynamic behavior and the correlation of parameter changes with operational constraints. This would enhance both immediate system performance and long-term reliability. Fig. 1 shows a high-level block diagram for boundary-based control is founded upon the basis of hybridized estimation.

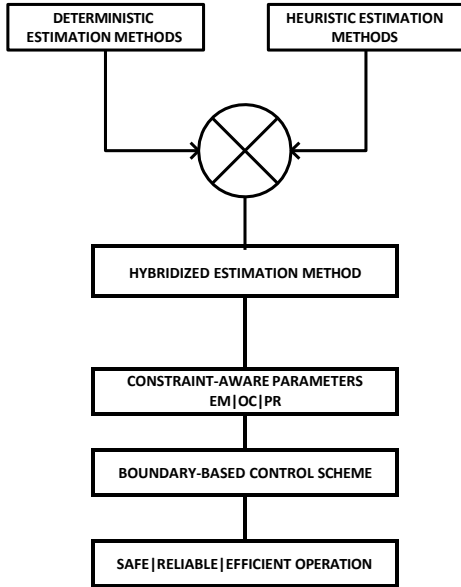


Figure 1. High-level block diagram of the hybridized estimation framework integrated with boundary-based control. Deterministic and heuristic estimation layers produce constraint-aware parameters for Efficiency Maximization (EM), Optimized Capability (OC), and Protection Resilience (PR), which drive adaptive control strategies for safe, reliable, and efficient operation.

III. PARAMETER ESTIMATION & OPERATIONAL CONSTRAINTS

Efficiency, capability, protection, and stability are critical to synchronous machine operation. Loss of efficiency raises operating costs and disrupts economic dispatch. Capability is set by thermal and power limit, such that exceeding them risks overload, thermal runaway, and damage. Operating outside the rated power factor increases I^2R losses, accelerates heating, and can cause insulation failure [24], with prolonged deviations worsening the damage [6]. Protection and stability are equally vital. While efficiency and capability drift gradually, faults are immediate and catastrophic [25]–[26], and can trigger magnetic saturation, system instability, load shedding, or voltage collapse.

Efficiency, capability, and protection/stability must be satisfied for safe, reliable, and economical operation.

Expressing these constraints in measurable parameters builds a behavior-based link between parameter changes and operational limits, with each constraint’s boundary defining the maximum permissible deviation. Since these boundaries can be quantified, Efficiency Maximization (EM), Optimized Capability (OC), and Protection Resilience (PR) can be directly embedded into synchronous machine parameter estimation, forming the basis for constraint-aware, behavior-based methods suited for advanced control and protection.

In this work, *constraint-aware* means that estimated parameters are explicitly linked to measurable operational limits in efficiency, capability, and protection margins. *Behavior-based* means that estimation reflects dynamic machine responses to disturbances rather than static fits to offline tests.

A. Efficiency Maximization (EM)

Undoubtedly, the parameters of a synchronous machine are the macroscopic elements that shape its characteristics. Critical parameters such as reactance and resistance values, directly influence the quota of hysteresis, eddy current and anomalous losses that a system yields. Based on the parameters of a machine and its operating scenarios, there exists a maximum efficiency threshold. This threshold corresponds to the value of the design efficiency of the machine [27].

As such Efficiency Maximization (EM) can be defined as the condition in which a rotating electrical machine operates as close as possible to its design-rated efficiency, despite the presence of dynamic, ferromagnetic, electrical, and mechanical losses. This condition reflects the machine’s ability to maintain minimal performance degradation under dynamic loading and disturbance conditions.

B. Optimized Capability (OC)

A synchronous generator converts mechanical input to active (P) and reactive (Q) power, with maximum values limited by field, armature, and core end heating constraints. Exceeding these risks overheating and damage [24]. Power capability curves [28]–[30] define the safe P–Q range, with the operating point, set by demand and excitation—tracing a trajectory within these limits [6]. Maximizing output under varying loads requires keeping operation in regions where rated power can be sustained without breaching thermal limits, controlling response to maintain safe margins [31], [25]. Power factor, inertia, reactance, and subtransient reactance all influence dynamic response and optimum capability.

Optimized Capability (OC) can be defined as the condition in which a rotating electrical machine delivers the maximum permissible real and reactive power while operating within the constraints set by its power capability, thermal limits, and damage thresholds.

C. Protection Resilience (PR)

For synchronous generators, protection and control rely on parameters such as current, voltage, frequency, speed, power, and temperature. Relays with instrument transformers and sensors trip breakers when thresholds are exceeded [32].

Settings are typically entered manually and updated with system changes, but operational drift can reduce accuracy and resilience [33], [34]. Conventional relays have limited predictive ability, though methods such as Farantatos et al. [35] integrate stability monitoring with out-of-step protection for anticipatory action. Disturbances affect rotor angle, frequency, and voltage stability, which are governed by generator and network parameters [36]–[38]. Zhu et al. [39] shows deep learning can forecast transient stability margins in real time, enabling adaptive settings. Parameters including damping, inertia, and transient/subtransient reactances determine disturbance response, short-circuit capacity, and reactive support [24], [25], [38], directly influencing protection and fault resilience [32].

Protection Resilience (PR) is therefore defined as the condition in which a protection scheme for a rotating electrical machine can anticipate, contain, and recover from disturbances or fault conditions that threaten its safety and stability.

D. A Significant Research Gap

Synchronous machine parameters govern efficiency, protection resilience, and capability—each defining operational boundaries. Parameter changes shift response toward or away from these limits, with breaches risking unsafe operation. Estimation must move beyond static fitting to capture dynamic, time-varying parameters under real-world constraints.

Conventional methods apply tests or disturbances to derive parameters [6] but cannot identify disturbance type from results [11], [42], limiting targeted action for EM, OC and PR. Adding interpretability is key to linking control with enhanced estimation. An improved approach should capture short- and long-term variations, treat parameters as time-varying, and integrate forecasting to track trajectories within EM, OC and PR for optimized control.

Heuristic and deterministic methods are not invariant to machine design or operation [43], such that changes require re-estimation. Hybrid approaches can address this, combining AI’s adaptability with deterministic accuracy to deliver real-time, behavior-aware parameters tied to efficiency maximization and protection resilience [12].

IV. BOUNDARY-BASED CONTROL

There is a need for a hybridized estimation technique capable of deriving behavior-based, constraint-aware parameters linked to the operational constraints of EM, OC, and PR, which capture the response of a synchronous machine to operational changes or disturbances [44], [45]. Each constraint can be expressed as a function of these parameters, defining a boundary or operational limit. Parameter variation produces a trajectory toward or away from an equilibrium point representing design operating conditions [46]. For a synchronous generator coupled to a network, disturbance response creates a unique trajectory within this constraint space, outlining a dynamic boundary governed by time-varying parameter estimates. Monitoring this trajectory in real

time enables assessment of proximity to operational limits and supports corrective action before they are exceeded [47].

By manipulating parameters within a common solution plane, all three constraints—EM, OC, and PR—can be simultaneously addressed. This enables derivation of an ideal operating point that unifies the constraints into a single control objective [48]. This operating point can be used as a setpoint within a nonlinear optimization control strategy, whose aim is to achieve safe, reliable, and optimal machine operation. In practice, this setpoint can be applied within a nonlinear optimization-based controller, such as model predictive control (MPC) or adaptive gain scheduling, allowing real-time tuning of excitation and governor inputs to maintain operation within EM, OC, and PR boundaries. Fig. 2 shows the intersection of EM, OC & PR to create the ideal operating region.

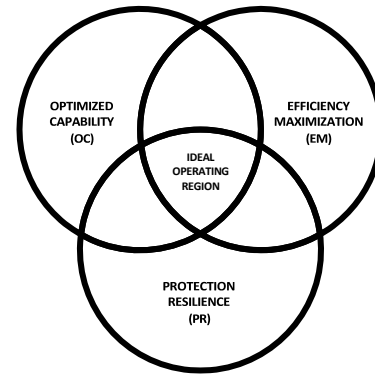


Figure 2. Conceptual representation of the three operational constraints: Efficiency Maximization (EM), Optimized Capability (OC), and Protection Resilience (PR). Their intersection defines the ideal operating region, where all constraints are satisfied simultaneously.

While parallels may be drawn with existing self-tuning control methods, the proposed concept is unique in that it simultaneously manipulates three control constraints within a non-linear optimization framework [45], [49]. Solving this optimization problem produces a setpoint that can be applied to a novel control algorithm. This approach creates an intelligent and adaptive control framework capable of continuously monitoring the power network and dynamically adjusting machine operation to satisfy the triple constraints in real time.

V. NEED FOR HYBRIDIZATION & BOUNDARY-BASED CONTROL SCHEMES

There is a clear need to bridge the gap between parameter estimation and the behavioral response of synchronous machines under EM, OC and PR constraints. As Industry 5.0 approaches, deterministic and heuristic methods must be hybridized to deliver both accurate physical parameters and real-time, constraint-aware estimates. Table 1 compares these approaches, underscoring hybridization as the first step toward an enhanced methodology.

This technique should integrate into a boundary-based intelligent control architecture, where EM, OC, and PR define a dynamic operational boundary normalized in a single solution plane. The convergence point represents safe,

reliable, and efficient operation, with hybridized estimation aimed at maintaining it in real time.

Berdin *et al.* [11] identify Newtonian and AI methods as highly accurate but lacking an automated, machine-invariant link between behavior-based estimation and boundary-based control. While most pre-2021 methods show limited promise, several works from 2013–2025 [13], [46]–[52] demonstrate progress in interpretability, machine invariance, and boundary-based capability—further motivating advancement toward integrated protection and control.

TABLE I. COMPARISON OF DETERMINISTIC, HEURISTIC, AND HYBRIDIZED ESTIMATION METHODS

Criteria	Deterministic	Heuristic (Data-Driven)	Hybridized
Accuracy	High if model and inputs are correct	Variable; depends on training data quality	High with improved robustness through dual physical–data validation
Interpretability	Fully transparent; grounded in physical laws	Often opaque (e.g., ANN, deep learning)	Partial to high, depending on AI architecture and use of physical constraints
Real-Time Capability	Limited (offline methods or slow for large models)	High once trained	High via pre-trained AI with dynamic tuning
Parameter Coverage	Moderate; covers static and dynamic parameters	Broad; captures nonlinear behavior	Broad with physical validation of behavior-based estimates
Machine Invariance	Low; requires re-estimation for each machine	Conditional; needs large, diverse datasets	High when trained on normalized structures with encoded metadata
Forecasting Capability	None; reactive only	Excellent (e.g., LSTM, Kalman filters)	Enabled via forecasting on surrogate markers
Operational Constraint Awareness	Rare; requires explicit integration	Rarely integrated	Actively embedded in estimation process
Cost–Benefit Potential	Limited scalability; high per-machine setup cost	Moderate; training cost can be high	High; scalable deployment, reduced tuning effort, extended asset life

VI. CONCLUSION

This work demonstrates that hybridizing deterministic accuracy with heuristic adaptability offers a decisive advantage over single-method approaches for synchronous machine parameter estimation. By embedding EM, OC, and PR constraints into the estimation process, the proposed method produces constraint-aware parameters that correlate directly with real-time machine behavior. These parameters define dynamic operational boundaries, enabling boundary-based control to maintain safe, reliable, and efficient operation

under changing conditions. The framework is machine-invariant, supports forecasting, and scales effectively across diverse installations, reducing downtime and integration costs while improving asset life. Beyond immediate operational gains, this approach establishes a clear pathway to integration with advanced adaptive control strategies and digital twin platforms, positioning it as a cornerstone for future Industry 5.0-ready power systems.

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